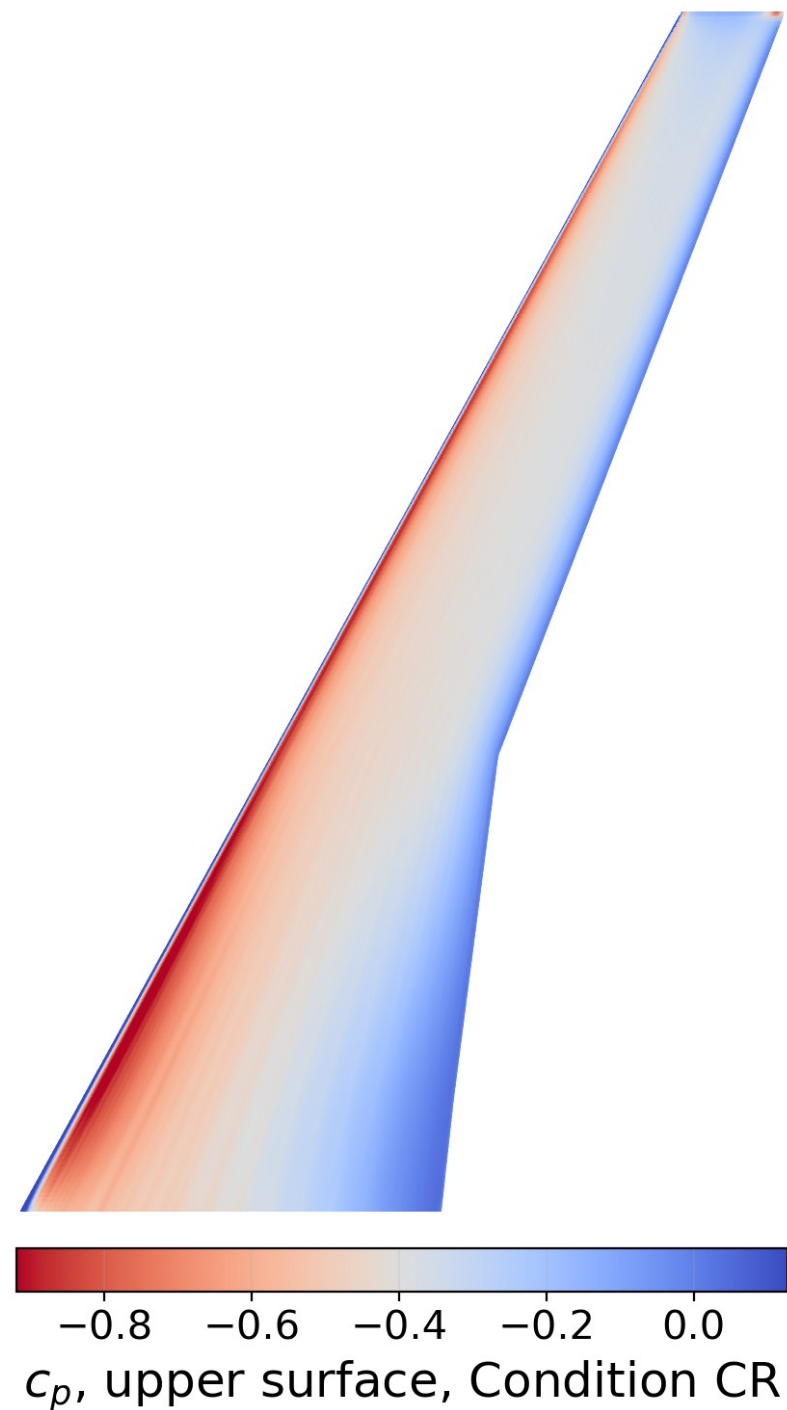


# ARGUS morphing wing: RANS validation of the VLM design study

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9 October 2026



# My job on ARGUS, and the three questions

- Validate the VLM candidate selection against viscous RANS on six wings.
- Six geometries, three operating points, 90 converged cases, 18 matched-lift trims.
- Questions: what was done, what every quantity gives, is the VLM trustworthy.
- Early and late cruise on top, which the VLM does not have.

# Operating points and OpenFOAM setup

- Reference area  $S_{\text{ref}}$  1.24092 m<sup>2</sup> and chord  $C_{\text{ref}}$  0.39396 m, full wing.
- RANS solves the half wing on  $S_{\text{ref}}/2$ ; one drag count is  $10^{-4}$ .

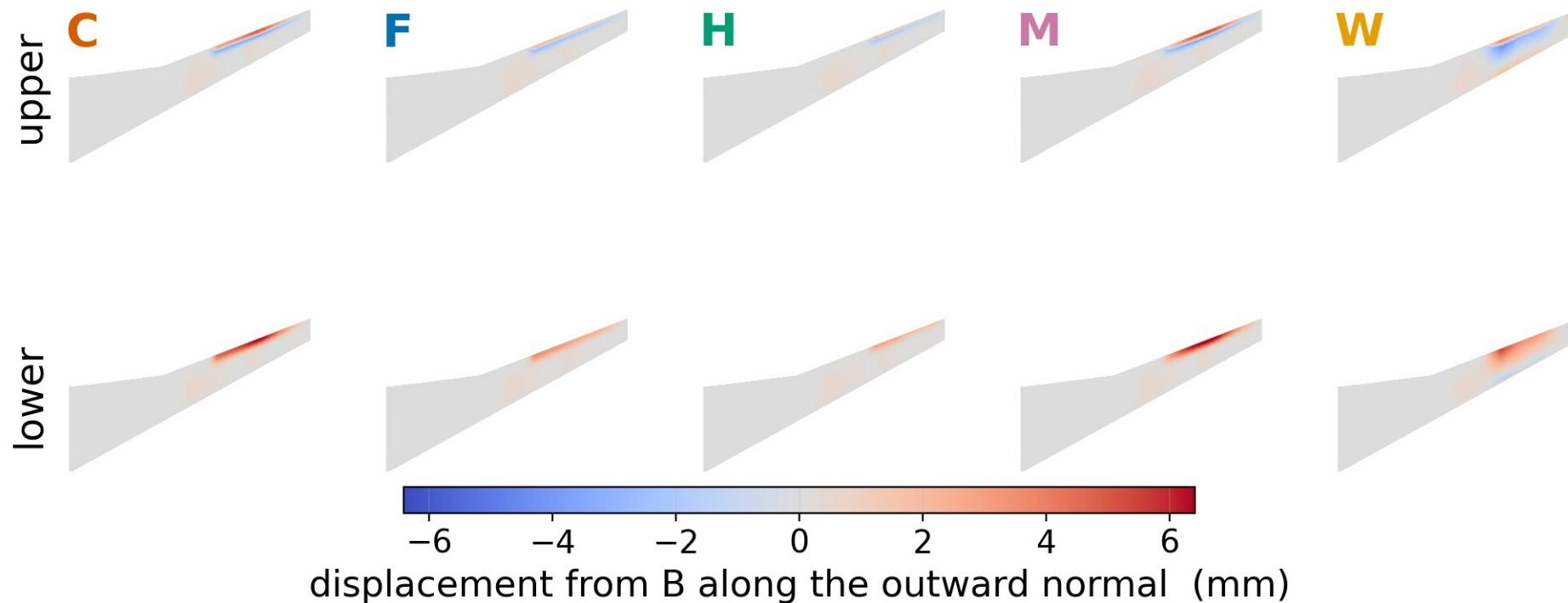
|                                    | Condition CR                           | Early cruise                          | Late cruise                           | VLM, Condition CR        |
|------------------------------------|----------------------------------------|---------------------------------------|---------------------------------------|--------------------------|
| Mach                               | 0.10                                   | 0.78                                  | 0.78                                  | 0.10                     |
| Target $C_L$                       | 0.428277635108                         | 0.529297087450                        | 0.544114800210                        | 0.428277635108           |
| $U_\infty$ (m/s)                   | 34.0                                   | 233.9                                 | 230.5                                 | n/a                      |
| Re on $C_{\text{ref}}$             | $0.92 \times 10^6$                     | $2.1176454 \times 10^7$               | $1.6107442 \times 10^7$               | $1.0 \times 10^7$ (deck) |
| Solver                             | incompressible,<br>incompressibleFluid | compressible, fluid<br>transonic      | compressible, fluid<br>transonic      | vortex lattice           |
| Turbulence model                   | kOmegaSST                              | Spalart-Allmaras                      | Spalart-Allmaras                      | none                     |
| Wall treatment                     | resolved,<br>nutLowReWallFunction      | modelled,<br>nutUSpaldingWallFunction | modelled,<br>nutUSpaldingWallFunction | none                     |
| First-cell $y^+$ , top /<br>centre | 0.541 / 0.270                          | 80.3 / 40.1                           | 80.3 / 40.1                           | n/a                      |
| Prism layers                       | 11, first 4.5 $\mu\text{m}$ , r 1.450  | 5, first 43.7 $\mu\text{m}$ , r 1.300 | 5, first 43.7 $\mu\text{m}$ , r 1.300 | n/a                      |
| Surface cell                       | level 11, 0.488 mm                     | level 11, 0.488 mm                    | level 11, 0.488 mm                    | n/a                      |

# The six geometries

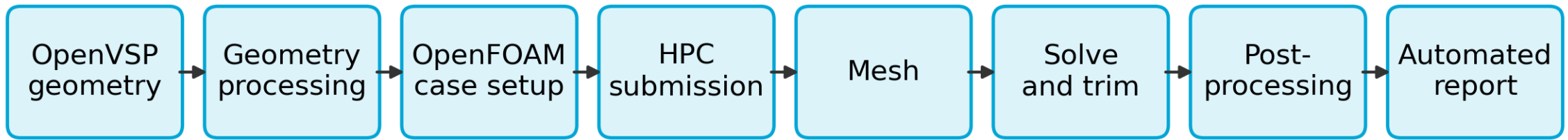
- Three mechanisms, five candidates, one rigid baseline B.
- Trailing edge moves 3 to 7 mm over the outer 39% span.

| ID | DSO case      | Concept              | What moves                    | Span, $\eta$   | Peak TE (mm) |
|----|---------------|----------------------|-------------------------------|----------------|--------------|
| C  | cte_i002_c04  | trailing-edge camber | aft chord from x/c 0.62       | 0.6125 to tip  | 6.89         |
| F  | cfft_b02_c01  | trailing-edge camber | aft chord from x/c 0.62       | 0.6125 to tip  | 3.45         |
| H  | chc_g02_c06   | hinged flap          | flap aft of x/c 0.70          | 0.710 to 0.970 | 3.15         |
| M  | mcv2_i002_c01 | trailing-edge camber | aft chord from x/c 0.62       | 0.6125 to tip  | 7.29         |
| W  | cffw_b01_c01  | distributed twist    | whole section, about x/c 0.25 | 0.6125 to tip  | 5.72         |

# Where each concept moves the surface

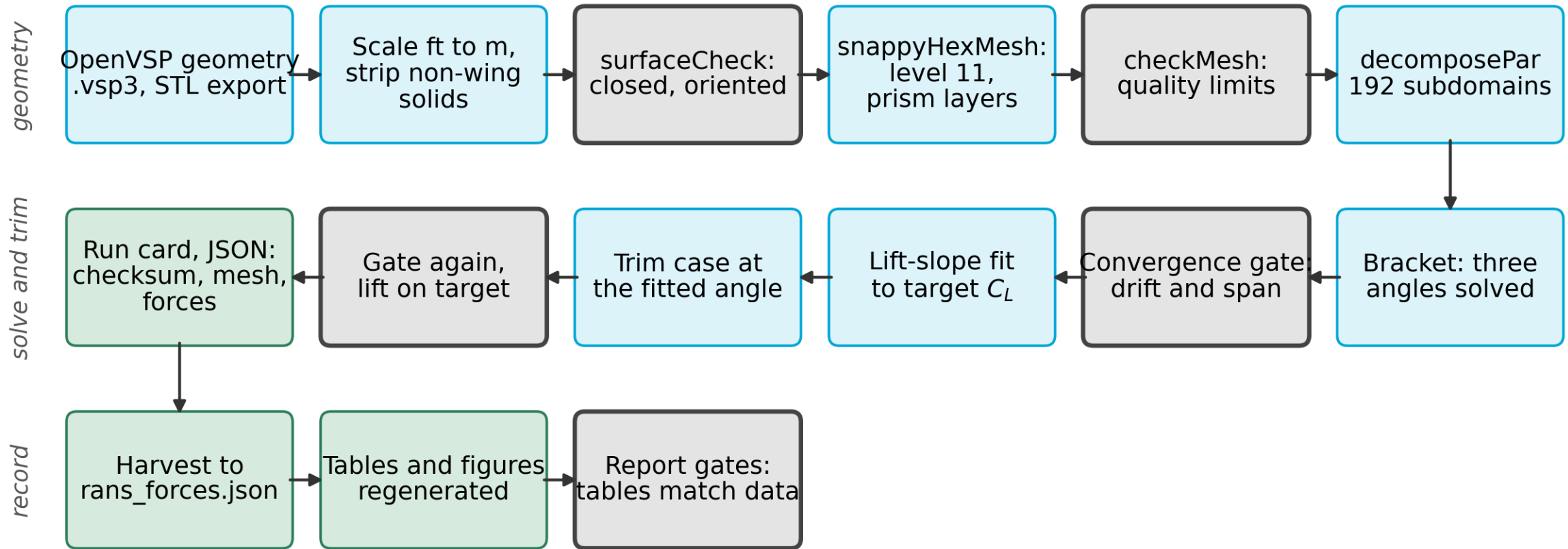


# The pipeline, end to end



- Geometry: scale feet to metres, strip non-wing solids, close the cap.
- Every case: mesh, three-angle bracket, trim, gate, run card.
- Every table and figure regenerates by script from one harvested file.

# The automated pipeline in detail



- Every geometry flies at one lift; drag at different lift compares nothing.

# Quantities of interest: the contents of the results

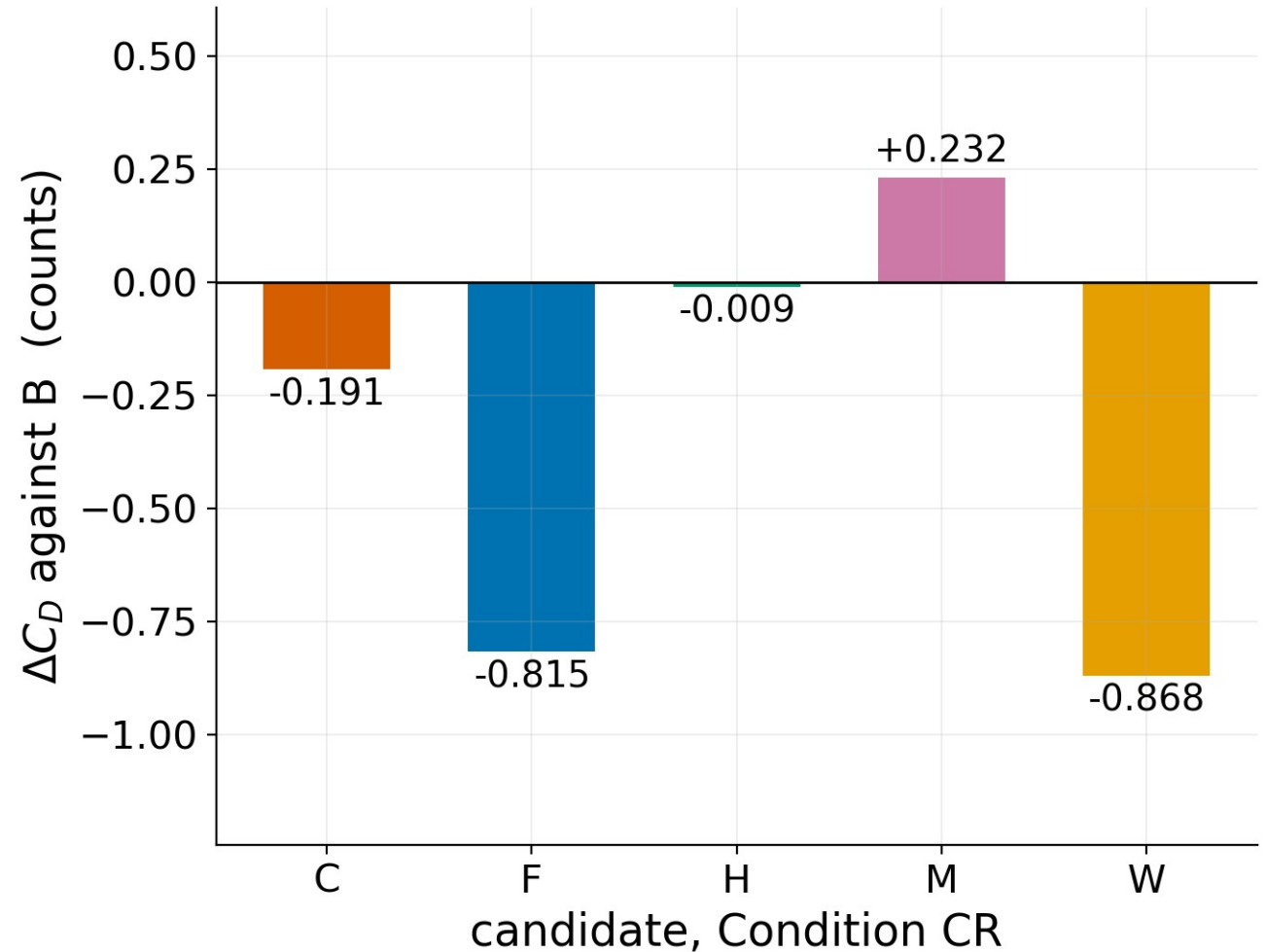
| Quantity                      | Definition                                                 | VLM (slide)       | RANS (slide)          | Baseline, VLM               | Baseline, RANS |
|-------------------------------|------------------------------------------------------------|-------------------|-----------------------|-----------------------------|----------------|
| Trimmed drag, each point      | candidate minus B at fixed $C_L$                           | induced only (22) | yes, 3 points (9, 10) |                             |                |
| Trim angle                    | incidence at the target $C_L$                              | yes (12)          | yes (12)              | 2.000°                      | 1.748°         |
| Spanwise load                 | $c_l c / C_{ref}$ per strip                                | yes (13)          | yes (13)              |                             |                |
| Induced drag, span efficiency | Trefftz integral and e                                     | near and far (14) | yes (14)              | 57.65 ct far,<br>91.68 near | 55.98 ct       |
| Hinge moment                  | moment about $x/c$ 0.62, $\eta \geq 0.60$ ; H, W own lines | no                | yes (15, 16)          |                             |                |
| Root bending                  | solver moment at the root                                  | yes (17)          | yes (17)              | 137.8 N m                   | 137.1 N m      |
| Surface $c_p$ and $c_f$       | sections and surface maps                                  | no                | yes (11, 18, 19)      |                             |                |



# Trimmed drag, Condition CR

- The whole field spans 1.1 counts around the baseline.
- Four of five candidates sit below the baseline drag.
- Read the low-speed sequence as a group, not a ranking.
- Every candidate trims at lower incidence than the baseline.

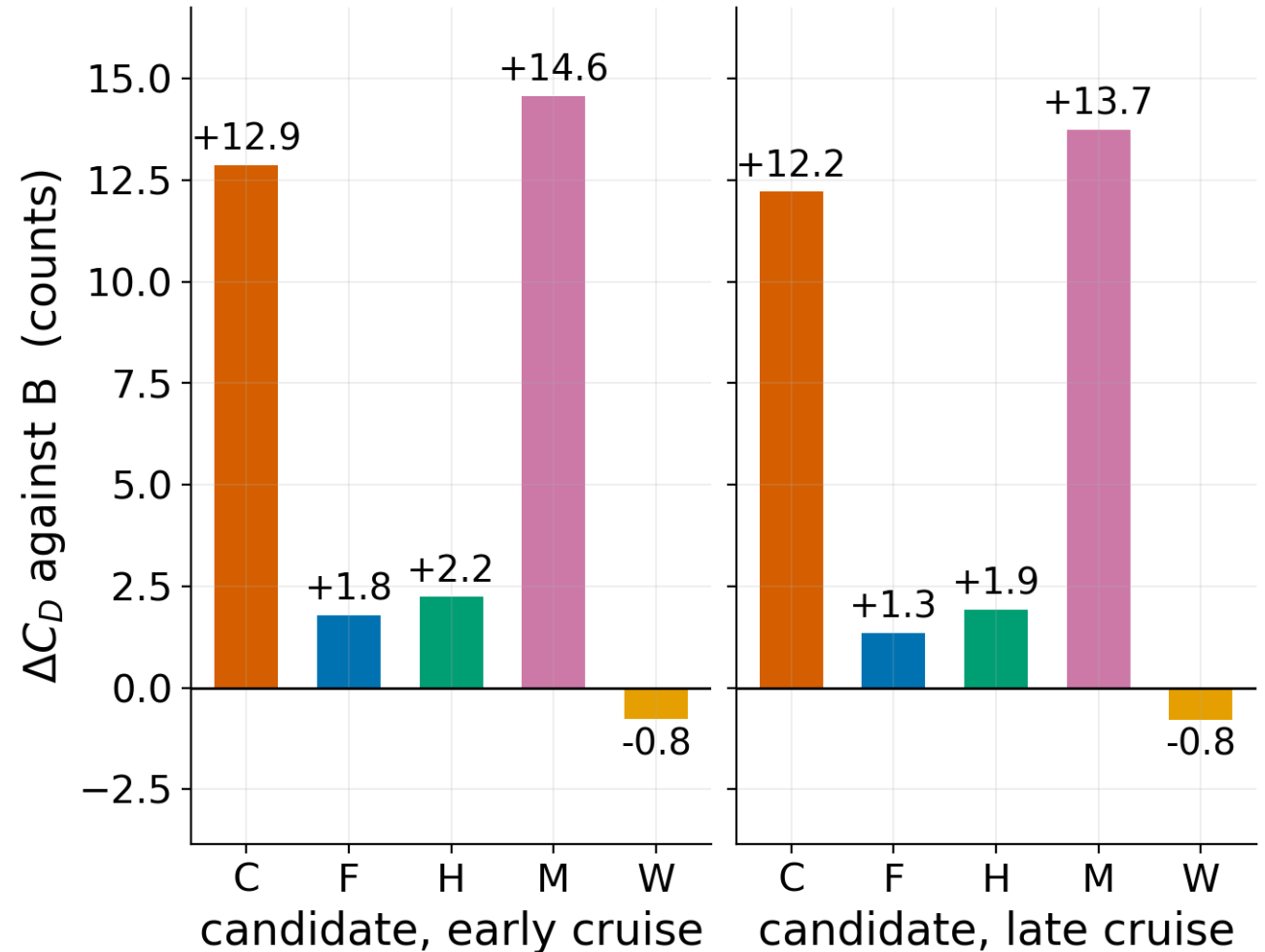
| ID | $C_D$ (ct) | $\Delta C_D$ (ct) | L/D   |
|----|------------|-------------------|-------|
| B  | 188.031    | +0.000            | 22.78 |
| C  | 187.840    | -0.191            | 22.80 |
| F  | 187.216    | -0.815            | 22.88 |
| H  | 188.022    | -0.009            | 22.78 |
| M  | 188.263    | +0.232            | 22.75 |
| W  | 187.163    | -0.868            | 22.88 |



# Trimmed drag, early and late cruise

- W saves drag at both cruise points, 0.77 and 0.78 counts.
- Both cruise points give the same order: W, F, H, C, M.
- C, M cost 12 to 15 counts; F, H about 2.
- Low-speed order differs only in the H and C pair.

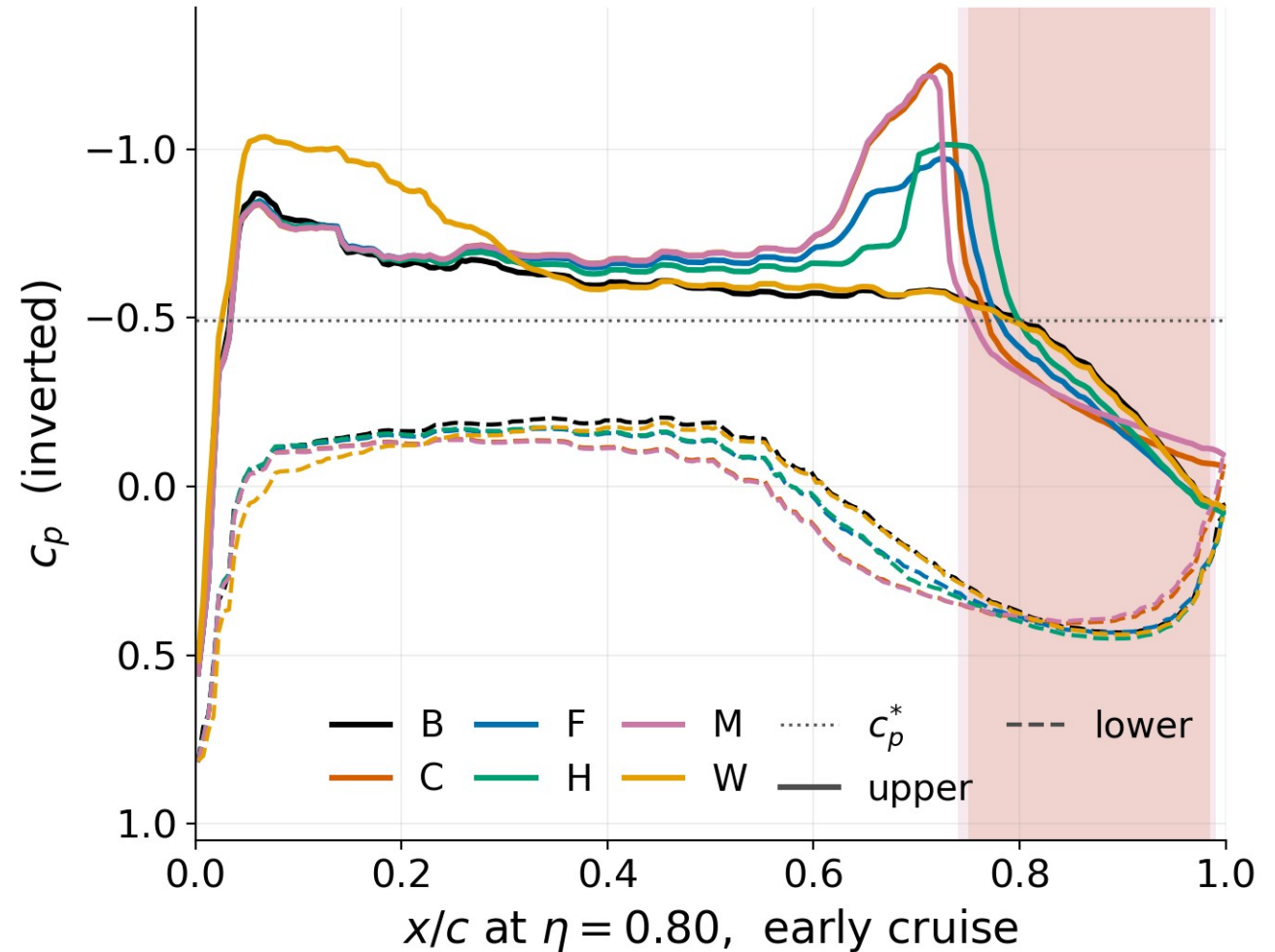
| ID | $\Delta C_D$ early (ct) | $\Delta C_D$ late (ct) |
|----|-------------------------|------------------------|
| B  | +0.000                  | +0.000                 |
| C  | +12.871                 | +12.214                |
| F  | +1.796                  | +1.348                 |
| H  | +2.244                  | +1.927                 |
| M  | +14.554                 | +13.728                |
| W  | -0.769                  | -0.781                 |



# Cruise drag follows the aft shock

- B and W: the supersonic region ends without a shock.
- F and H: a moderate shock, flow stays attached.
- C and M: a strong shock and separation behind it.
- Drag follows shock strength; late cruise repeats it.

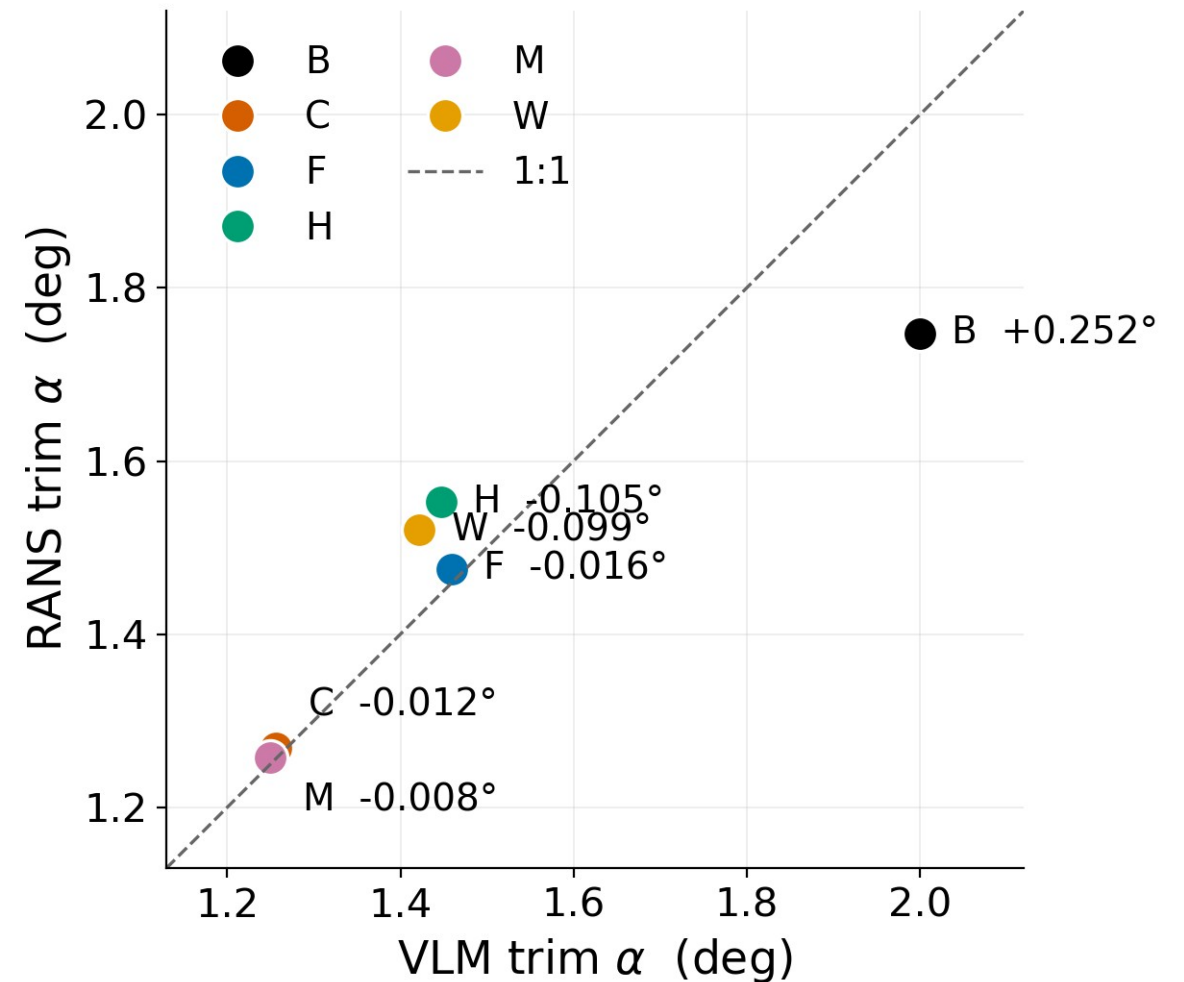
| ID | Shock $x/c$ | Peak $dc_p/d(x/c)$ | Reversed run, $x/c$ |
|----|-------------|--------------------|---------------------|
| B  | none        |                    | attached            |
| C  | 0.738       | 46                 | 0.75 to 0.98        |
| F  | 0.752       | 17                 | attached            |
| H  | 0.772       | 19                 | attached            |
| M  | 0.728       | 51                 | 0.74 to 0.99        |
| W  | none        |                    | attached            |



# Trim angle at the Condition CR target lift

- VLM matches the RANS trim angle within 0.11 degrees on every candidate.
- The baseline deck is the outlier, +0.25 degrees from the RANS.
- Same target lift on both sides, so no conversion enters.

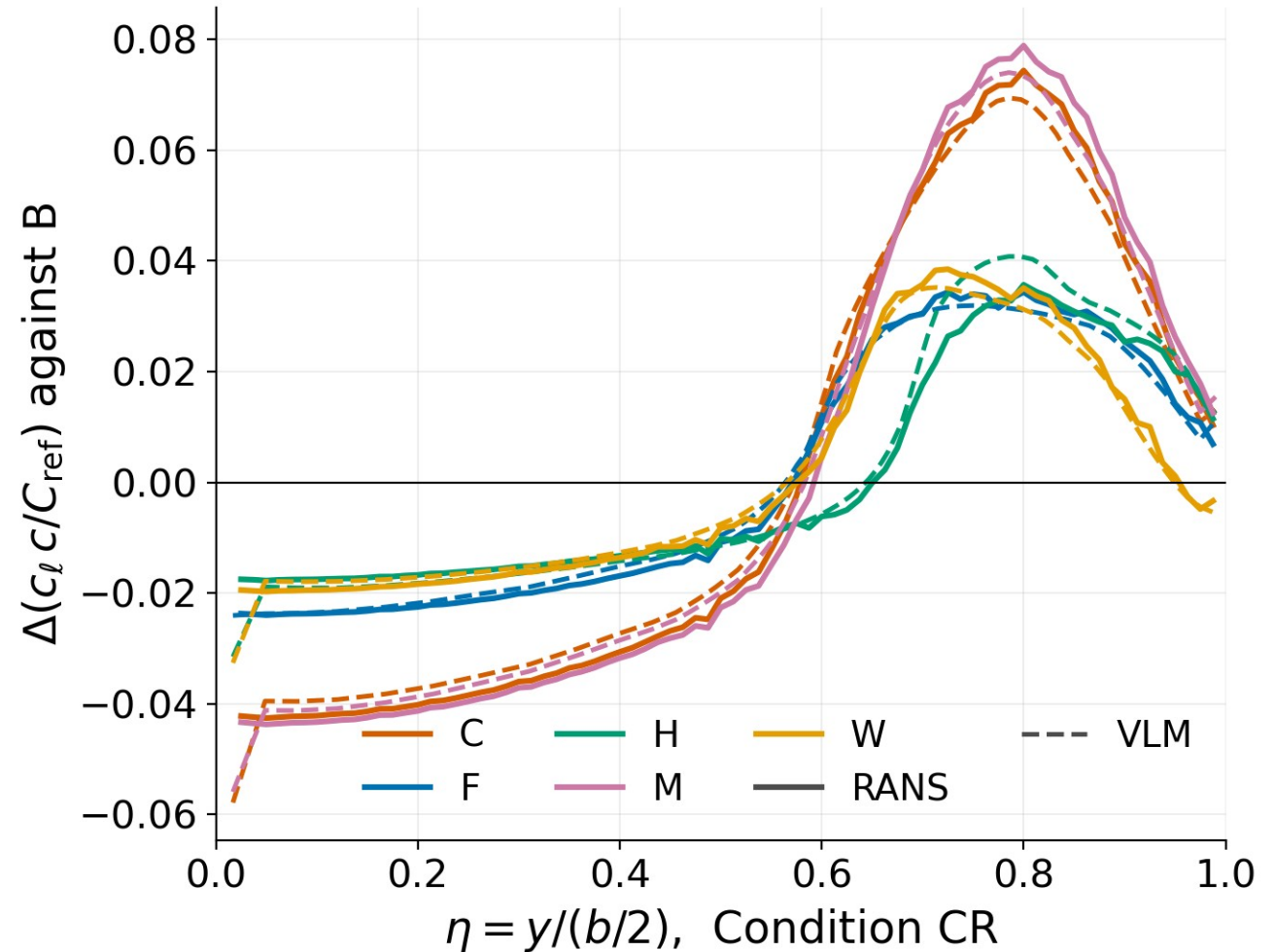
| ID | $\alpha$ RANS (deg) | $\alpha$ VLM (deg) | $\Delta\alpha$ (deg) |
|----|---------------------|--------------------|----------------------|
| B  | 1.7476              | 1.9998             | +0.2522              |
| C  | 1.2683              | 1.2565             | -0.0118              |
| F  | 1.4755              | 1.4594             | -0.0161              |
| H  | 1.5523              | 1.4470             | -0.1053              |
| M  | 1.2574              | 1.2497             | -0.0077              |
| W  | 1.5205              | 1.4219             | -0.0986              |



# Spanwise load, Condition CR

- The VLM puts the load change where the RANS puts it.
- R.m.s. difference 4.3 to 9.5 per cent of the peak change.
- The station of peak change agrees within 0.05 in  $\eta$ .

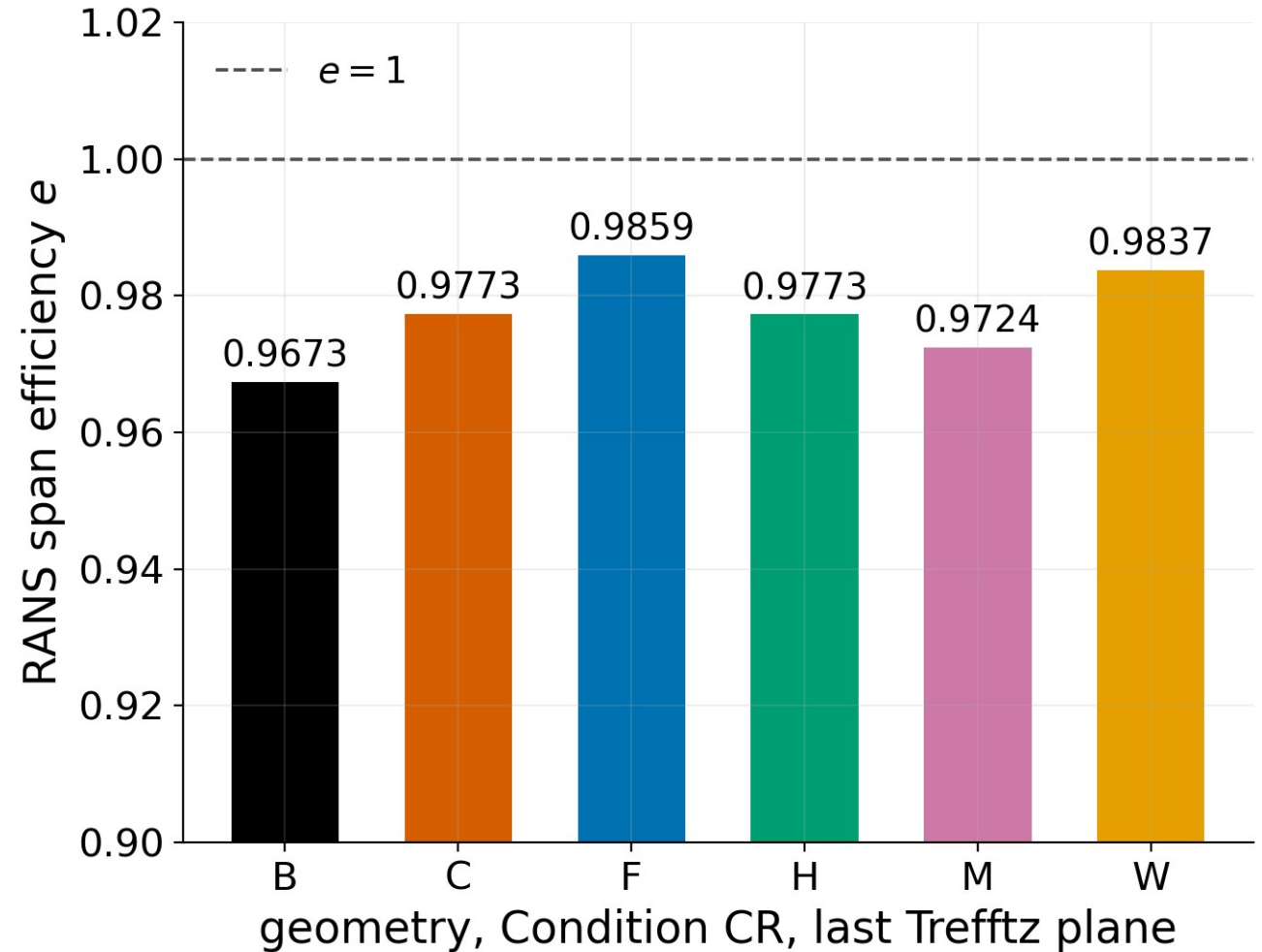
| ID | RANS peak $\Delta$ at $\eta$ | VLM peak $\Delta$ at $\eta$ | r.m.s., % of peak |
|----|------------------------------|-----------------------------|-------------------|
| C  | 0.074 at 0.80                | 0.069 at 0.79               | 4.6               |
| F  | 0.034 at 0.80                | 0.032 at 0.75               | 4.5               |
| H  | 0.036 at 0.80                | 0.041 at 0.79               | 9.5               |
| M  | 0.079 at 0.80                | 0.074 at 0.79               | 4.3               |
| W  | 0.039 at 0.72                | 0.035 at 0.71               | 5.5               |



# Induced drag and span efficiency, Condition CR

- Far-field VLM and RANS agree on the level, 0.94 to 0.99.
- Near-field VLM does not: 0.57 to 0.59 on the same runs.
- RANS  $e$  spans 0.9673 to 0.9859, every one below 1.

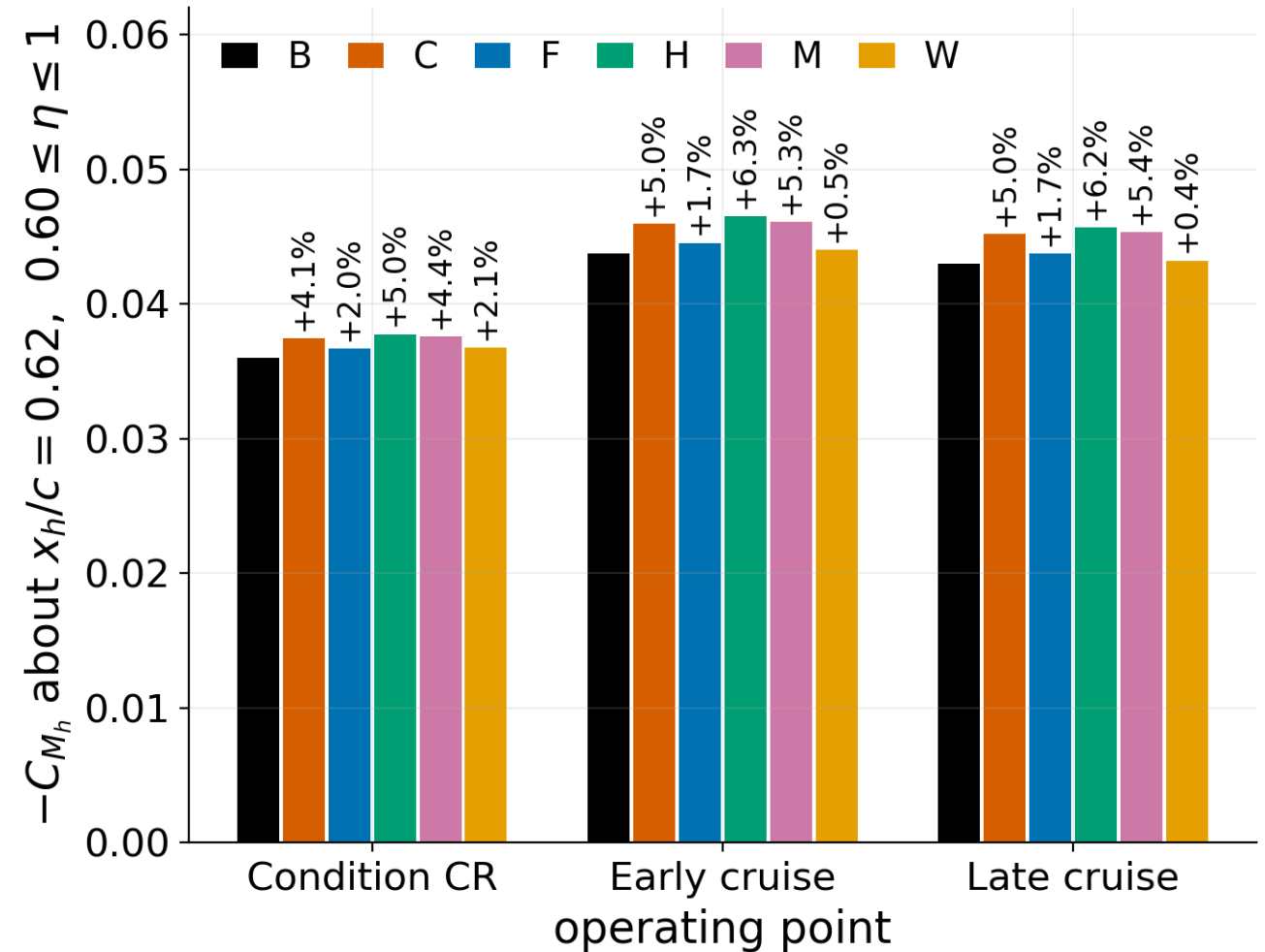
| ID | VLM E, near field (surface) | VLM E <sub>w</sub> , far field (wake) | RANS e, last plane |
|----|-----------------------------|---------------------------------------|--------------------|
| B  | 0.5907                      | 0.9394                                | 0.9673             |
| C  | 0.5865                      | 0.9627                                | 0.9773             |
| F  | 0.5733                      | 0.9721                                | 0.9859             |
| H  | 0.5679                      | 0.9605                                | 0.9773             |
| M  | 0.5847                      | 0.9595                                | 0.9724             |
| W  | 0.5699                      | 0.9701                                | 0.9837             |



# Hinge moment about $x/c = 0.62$ , outer span

- Every candidate raises the hinge moment, 0.4 to 6.3 per cent.
- The VLM cannot produce this quantity at all.
- H's and W's rows are comparisons, not their actuator loads.

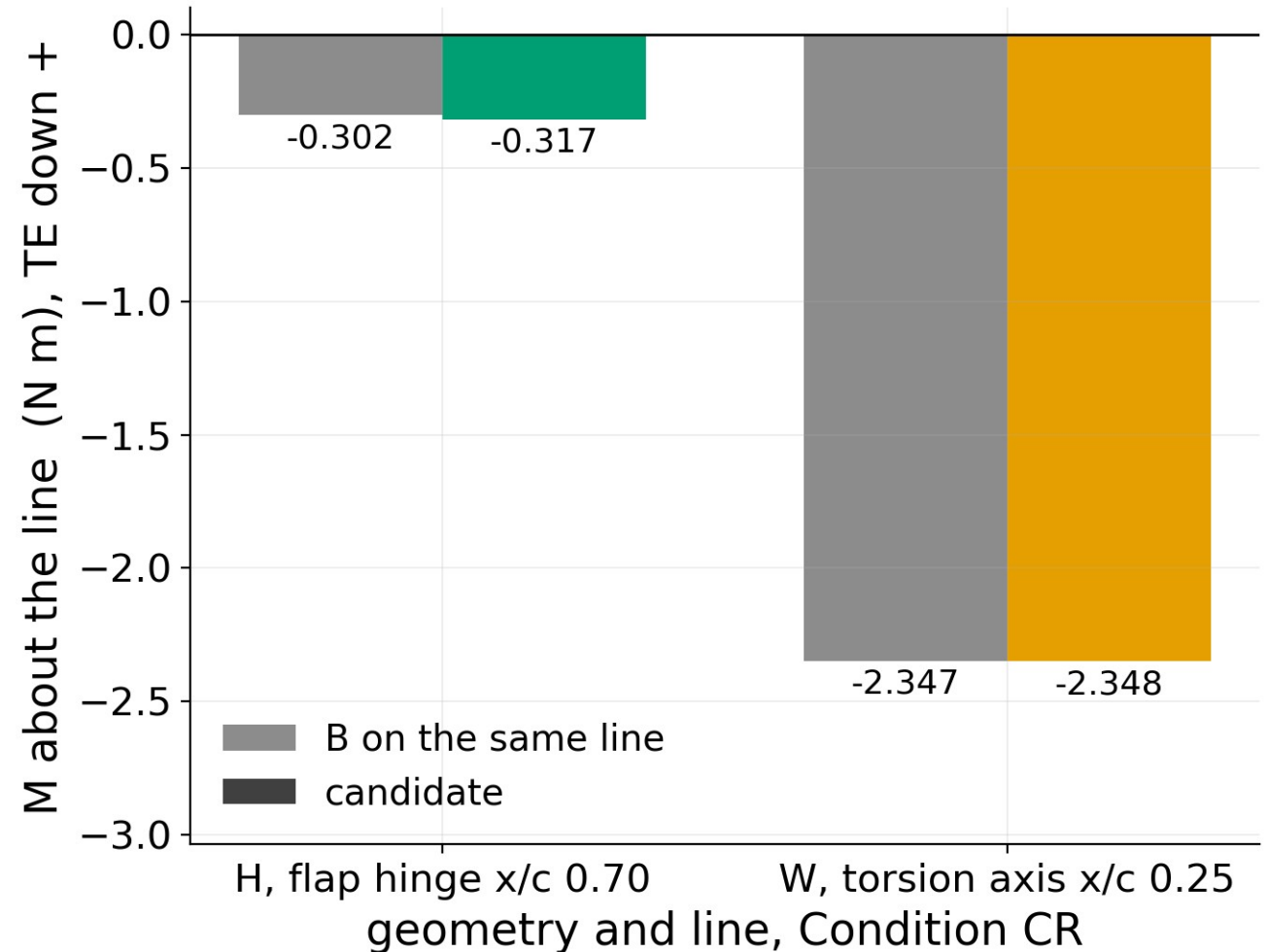
| ID | $\Delta$ %<br>Condition<br>CR | $\Delta$ % early<br>cruise | $\Delta$ % late<br>cruise | Actuator<br>load?       |
|----|-------------------------------|----------------------------|---------------------------|-------------------------|
| C  | +4.1                          | +5.0                       | +5.0                      | yes                     |
| F  | +2.0                          | +1.7                       | +1.7                      | yes                     |
| H  | +5.0                          | +6.3                       | +6.2                      | no, hinge at<br>0.70    |
| M  | +4.4                          | +5.3                       | +5.4                      | yes                     |
| W  | +2.1                          | +0.5                       | +0.4                      | no, twist<br>about 0.25 |



# Actuator loads on their own lines: H and W

- H's flap hinge at x/c 0.70: -0.317 N m, +5.1% against B.
- W's torsion about x/c 0.25: -2.35 N m, +0.0% against B.
- Current Condition CR evaluation, same gates and integrator as the common line.

| Geometry, line                    | M (N m) | B on that line | $\Delta  M $ % |
|-----------------------------------|---------|----------------|----------------|
| H, flap 0.70, $\eta$ 0.71 to 0.97 | -0.317  | -0.302         | +5.1           |
| W, torsion 0.25, $\eta$ 0.61 to 1 | -2.348  | -2.347         | +0.0           |

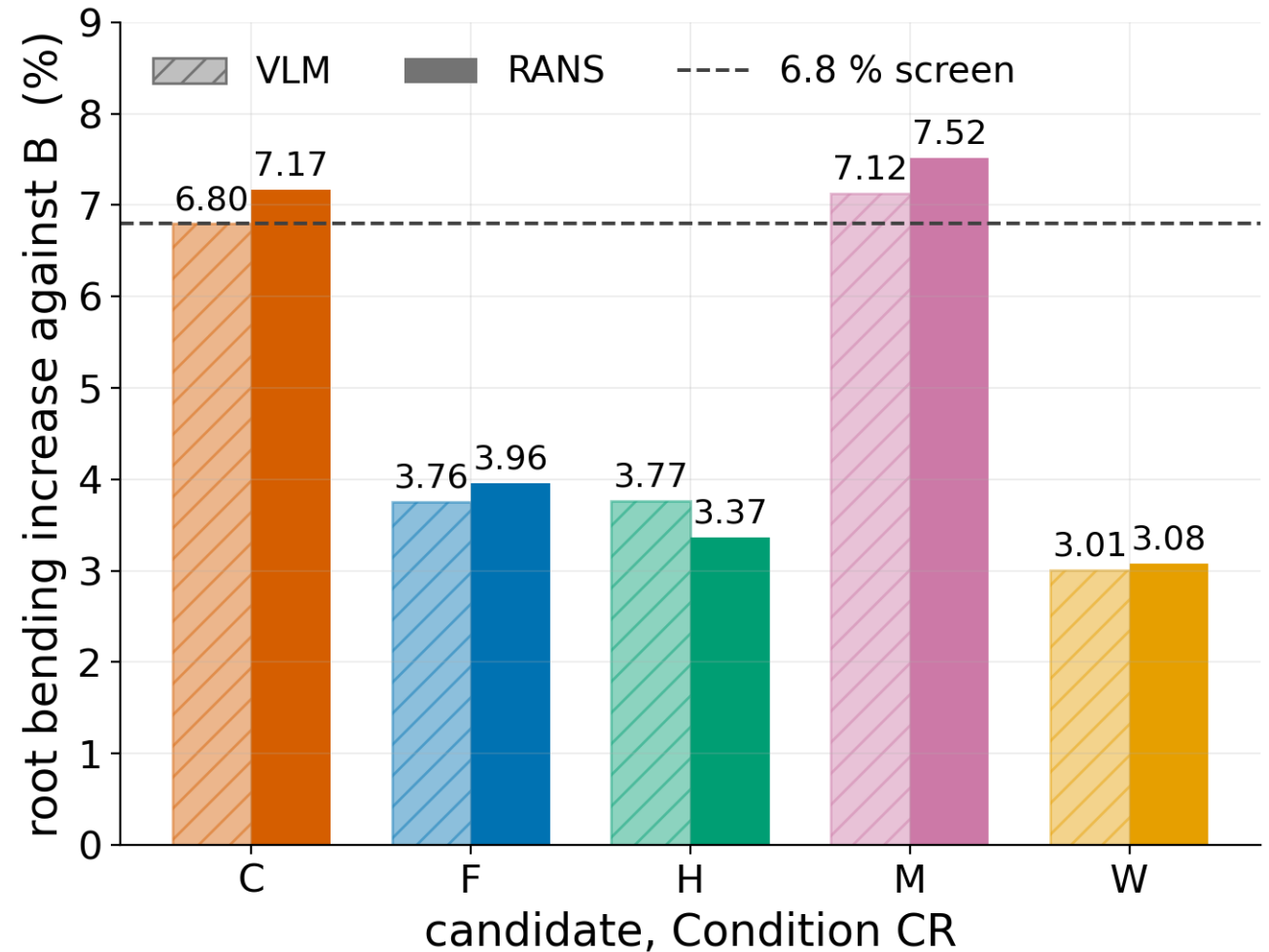




# Root bending moment, Condition CR

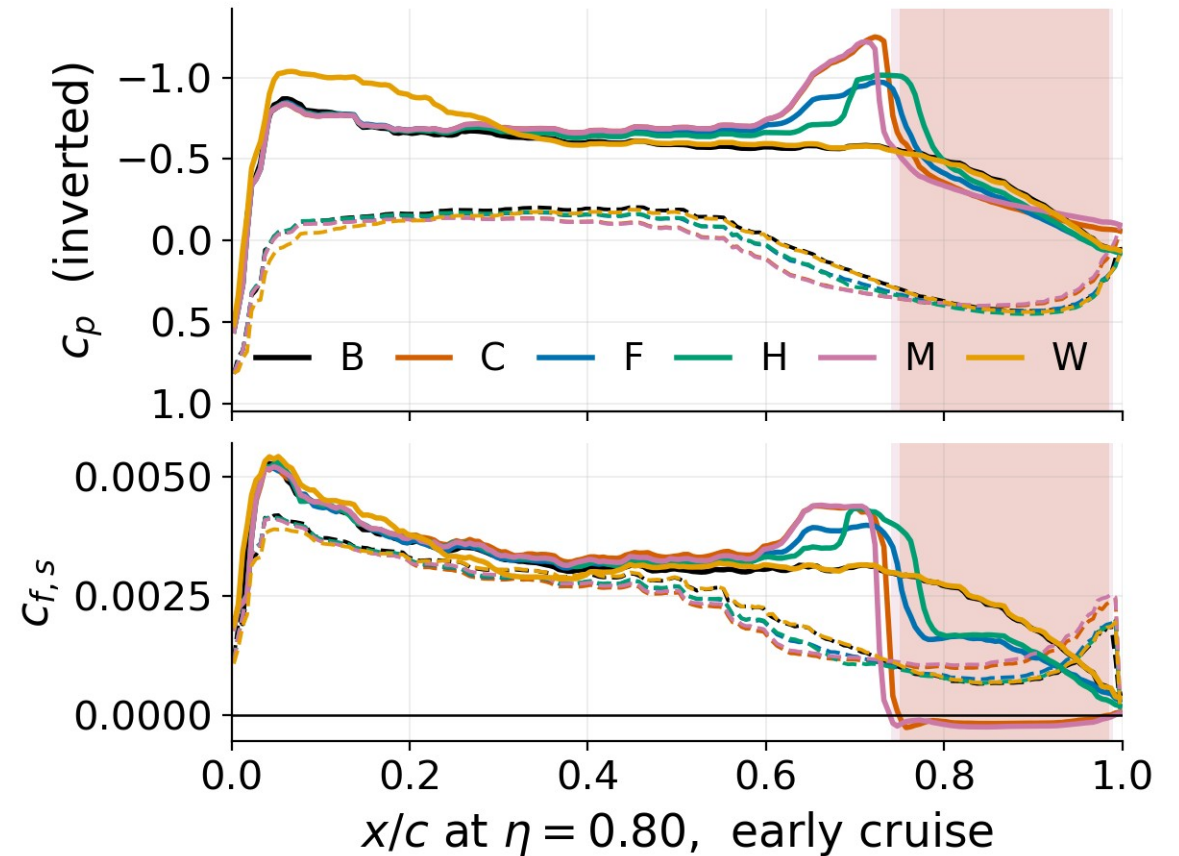
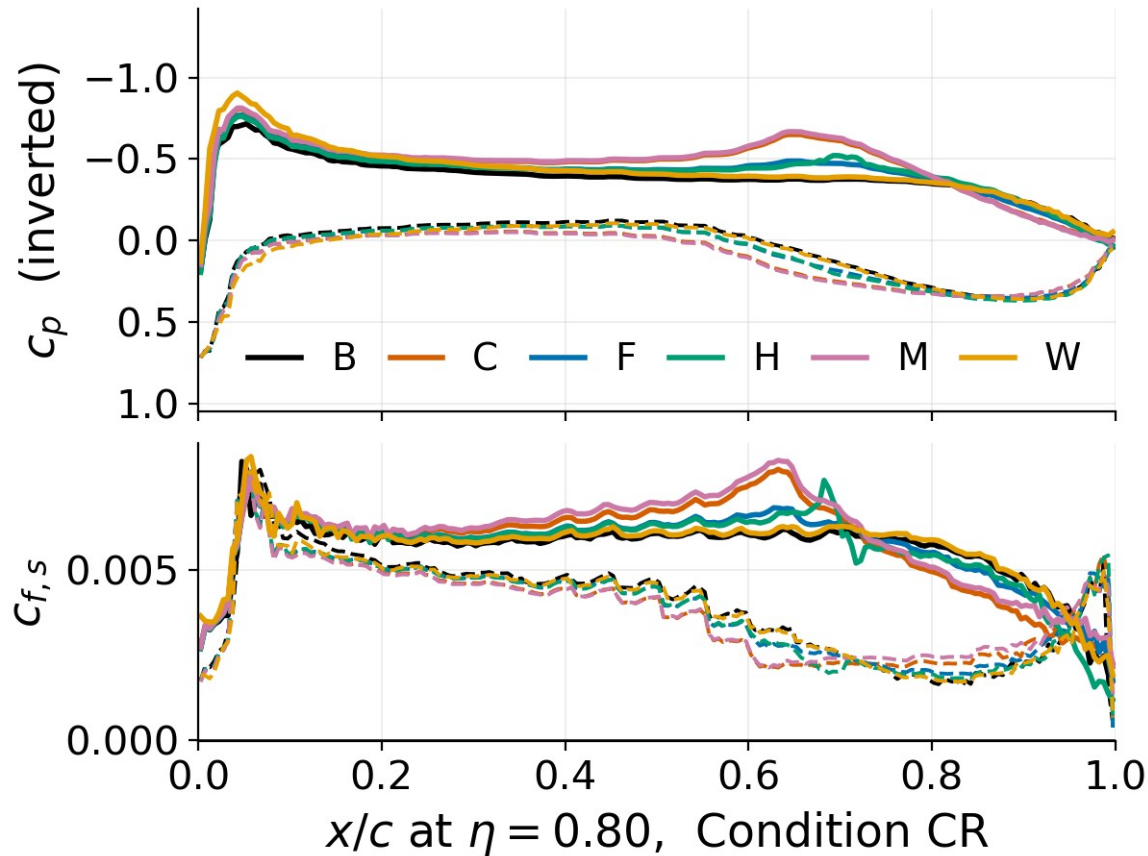
- VLM and RANS agree on root bending within 0.4 percentage points.
- C passes the 6.8% screen in the VLM, fails it in RANS.
- M exceeds the screen in both methods.

| ID | VLM %  | RANS % | Difference (pp) |
|----|--------|--------|-----------------|
| C  | +6.796 | +7.170 | +0.37           |
| F  | +3.755 | +3.960 | +0.20           |
| H  | +3.769 | +3.368 | -0.40           |
| M  | +7.124 | +7.519 | +0.40           |
| W  | +3.010 | +3.079 | +0.07           |



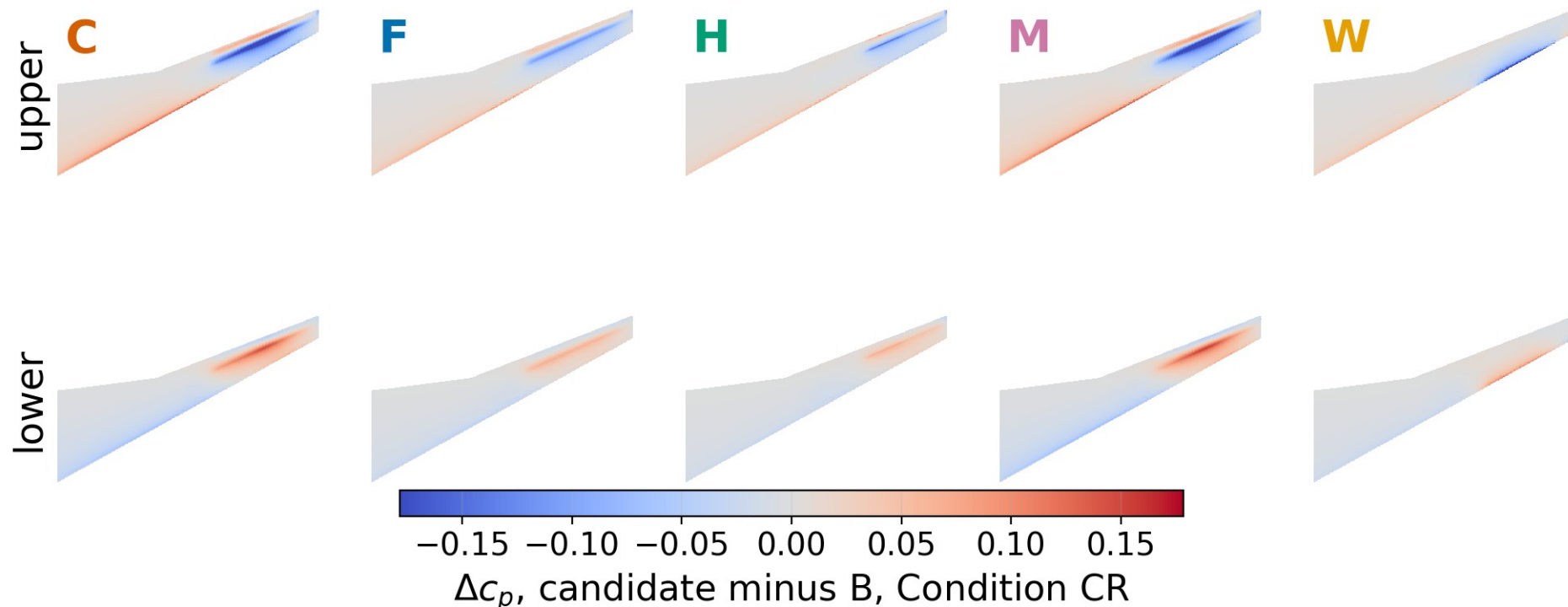
# Surface pressure and skin friction at $\eta = 0.80$

- Nothing separates at Condition CR: largest reversed upper fraction 0.02 per cent.
- Cruise,  $\eta$  0.80: C and M separate; F, H do not.



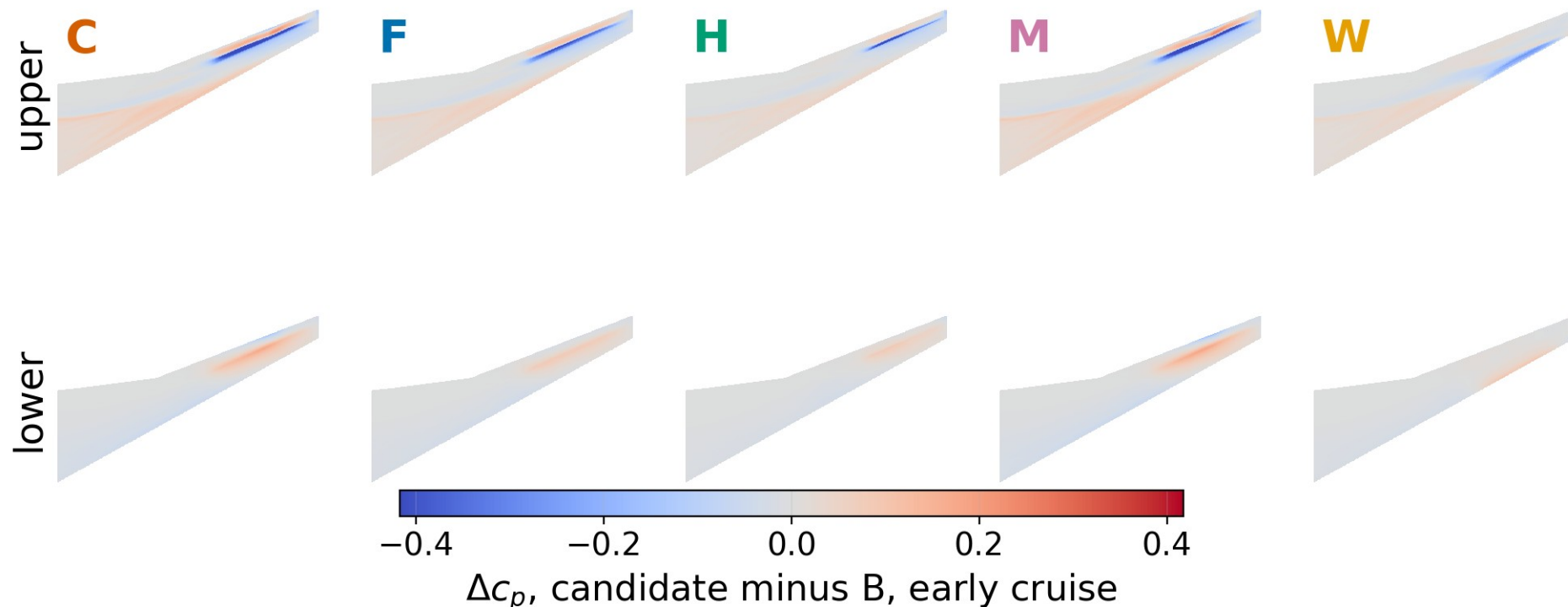
# Surface $\Delta c_p$ against the baseline, Condition CR

- Candidate minus B on one shared scale,  $\pm 0.18$  at the 99.5th percentile.
- Suction gain above and pressure gain below over the morphed outer span.



# Surface $\Delta c_p$ against the baseline, early cruise

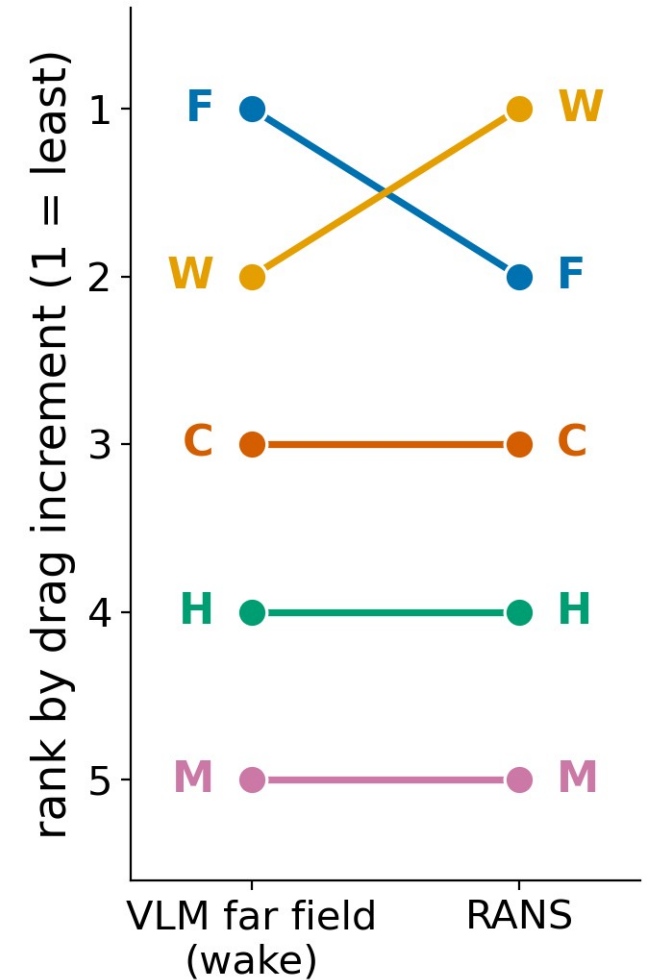
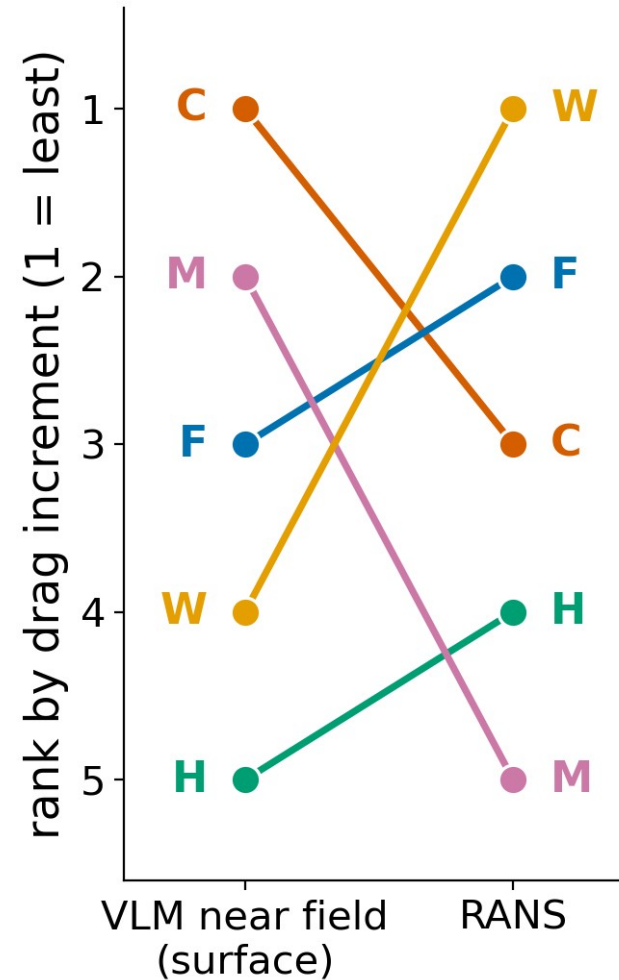
- Candidate minus B on one shared scale,  $\pm 0.42$  at the 99.5th percentile.
- Suction gain above and pressure gain below over the morphed outer span.



# Ordering: does the VLM rank the candidates?

- Far field reproduces the RANS order, 9 pairs of ten.
- Near field does not, 4 pairs of ten concordant.
- The one far-field error is the closest pair, 0.053 counts apart.

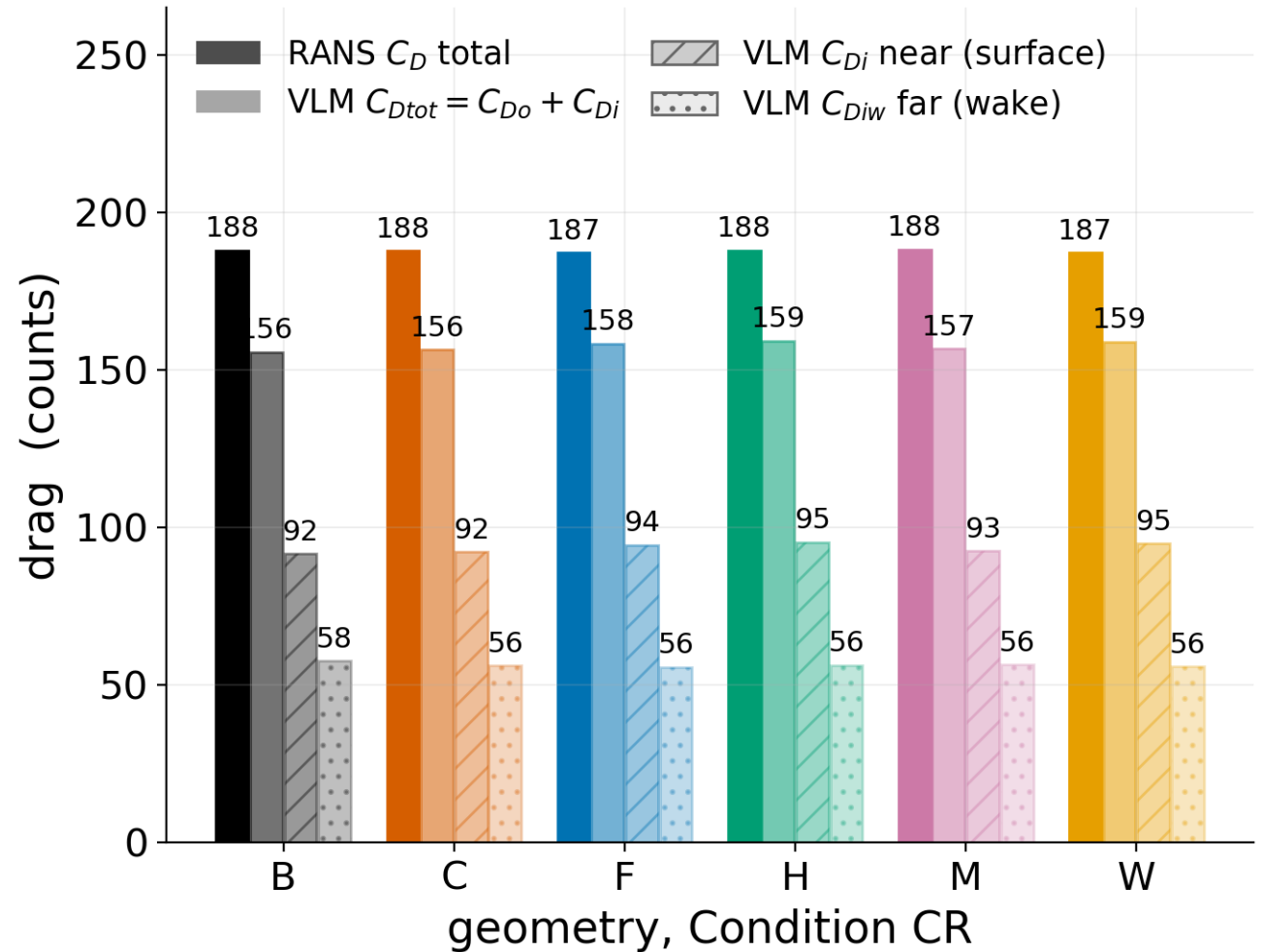
| ID             | RANS rank | Near field (surface) | Far field (wake) |
|----------------|-----------|----------------------|------------------|
| C              | 3         | 1                    | 3                |
| F              | 2         | 3                    | 1                |
| H              | 4         | 5                    | 4                |
| M              | 5         | 2                    | 5                |
| W              | 1         | 4                    | 2                |
| Kendall $\tau$ |           | -0.20                | +0.80            |



# Magnitude: does the VLM price the candidates?

- Far-field increments sit a near-constant 1.2 counts below the RANS increments.
- Remove that one offset and the candidates agree within 0.25 counts.
- Near-field errors span 0.7 to 4.2 counts, wrong sign on four.
- Absolute pricing needs that offset, which only RANS supplies.

| ID | RANS $\Delta C_D$ | Far $\Delta C_{Diw}$ | Far minus offset | Residual |
|----|-------------------|----------------------|------------------|----------|
| C  | -0.191            | -1.392               | -0.196           | -0.01    |
| F  | -0.815            | -1.938               | -0.743           | +0.07    |
| H  | -0.009            | -1.266               | -0.071           | -0.06    |
| M  | +0.232            | -1.209               | -0.014           | -0.25    |
| W  | -0.868            | -1.823               | -0.627           | +0.24    |



# Verdict

- The VLM orders candidates on its far-field column, 9 pairs of ten.
- Far-field increments hold to 0.25 counts after one common 1.2-count offset.
- At cruise W saves; C, M pay 12 to 15 counts.
- Every candidate raises the hinge moment; C fails the root-bending screen viscously.

# Thank you

Tyler Buchanan | [questions](#)



# Backup: cell counts

- Condition CR: 99.1 to 99.2 million cells; cruise: 73.6 to 73.8 million.

| ID | Concept              | Triangles | Condition CR (M) | Early cruise (M) | Late cruise (M) | Surface gate |
|----|----------------------|-----------|------------------|------------------|-----------------|--------------|
| B  | rigid baseline       | 192588    | 99.11            | 73.64            | 73.67           | pass         |
| C  | continuous TE camber | 192588    | 99.17            | 73.70            | 73.73           | pass         |
| M  | continuous TE camber | 192588    | 99.21            | 73.73            | 73.75           | pass         |
| F  | continuous TE camber | 195266    | 99.15            | 73.66            | 73.68           | pass         |
| W  | distributed twist    | 195266    | 99.20            | 73.67            | 73.69           | pass         |
| H  | conventional hinged  | 195266    | 99.12            | 73.67            | 73.69           | pass         |

# Backup: the convergence gate

- Verdict: CONVERGED, 55 of 55.

| Term      | Threshold       | Achieved over the 55 | Median |
|-----------|-----------------|----------------------|--------|
| Window W  | 200 samples     | 200 on all 55        | 200    |
| C_d drift | $\leq 0.05$ ct  | 0.0000 to 0.0130     | 0.0012 |
| C_l drift | $\leq 1.0$ ct   | 0.0004 to 0.2590     | 0.0615 |
| C_l span  | $\leq 1.0$ ct   | 0.0008 to 0.5374     | 0.1068 |
| Verdict   | all three terms | CONVERGED, 55 of 55  | n/a    |

# Backup: trim residuals

- Every trim sits within 0.90 counts of  $C_L$  of its target.

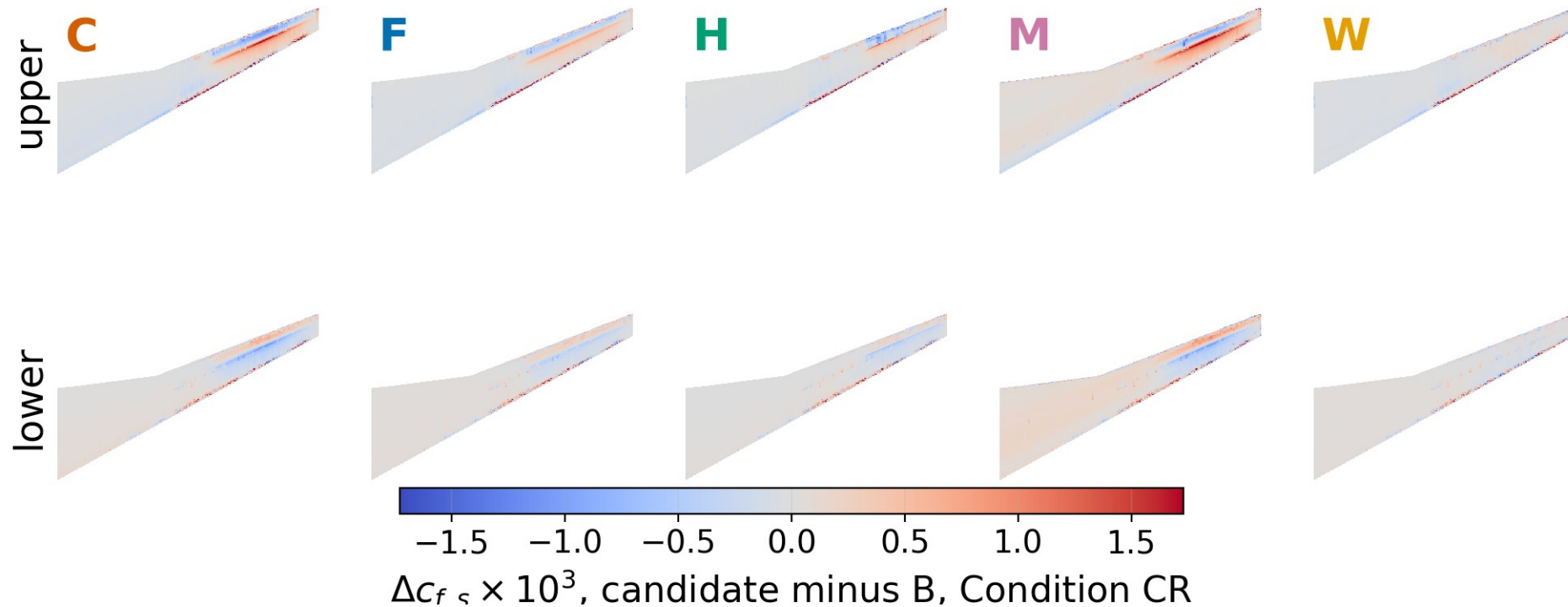
| ID | Condition CR | Early cruise | Late cruise |
|----|--------------|--------------|-------------|
| B  | -0.094       | +0.194       | +0.537      |
| C  | -0.005       | -0.305       | +0.752      |
| F  | -0.039       | -0.017       | -0.220      |
| H  | -0.039       | +0.087       | -0.372      |
| M  | +0.001       | +0.380       | +0.545      |
| W  | -0.037       | -0.565       | -0.895      |

# Backup: the wall-modelled question

- Surface cell 0.488 mm at level 11, identical for both conditions.
- Prism stack: 11 layers from 4.5  $\mu\text{m}$  against 5 from 43.7  $\mu\text{m}$ .
- Design  $y^+$  at the cell centre: 0.270 wall resolved, 40.1 wall modelled.
- That stack alone explains 99 to 99 versus 74 to 74 million.

# Backup: surface $\Delta c_{f,s}$ against B, Condition CR

- Candidate minus B,  $\pm 1.7 \times 10^{-3}$  at the 99.5th percentile.



# Backup: surface $\Delta c_{f,s}$ against B, early cruise

- Candidate minus B,  $\pm 1.8 \times 10^{-3}$  at the 99.5th percentile.

